

```
1 # Black Holes are Eggs
2
3 ## A speculative reproductive cosmology hypothesis
  with dark-sector gestation,
4
5 merged from a prior draft and an attached
  supplementary brief
6
7 **Prepared for Erik Olson**
8 **March 10, 2026**
9
10 **Status:** journal-style speculative manuscript
   grounded in standard GR/ $\Lambda$ CDM where established, with
   conjectural
11 interpretation isolated to the dark sector and black-
   hole interior dynamics.
12
13 > **Source integration note.** This manuscript
   explicitly incorporates the uploaded Grok brief *"
   Black Holes as Cosmic
14 Eggs: A Hypothesis for Reproductive Cosmology in a
   Living Universe"* as an auxiliary source [B0]. The
   merged version
15 > preserves that brief's Popławski-style torsion
   bounce, "Cosmic Fertility Index," and experimental
   program, while marking
16 > points of divergence where the combined superset
   becomes non-unique.
17
18 ---
19
20 # Abstract
21
22 We develop a speculative cosmological hypothesis in
   which the observable universe is treated as a living
   system embedded
23 in a wider reproductive ecology, and black holes
   function as eggs through which that system propagates
   lineage. The
24 external gravitational physics is kept standard:
   black holes remain Schwarzschild- or Kerr-like
```

24 compact objects to
25 current observational precision, and the cosmic
background remains close to flat Λ CDM. The
conjectural content enters in
26 the interpretation of the dark sector and of black-
hole interiors. Dark matter is allowed to include a
gestational
27 scaffold component that nourishes or stabilizes egg
formation, while dark energy is parameterized as an
order field that
28 may be interpreted, at the strongest metaphysical
reading, as a coordination or consciousness-like
field. A merged
29 fertility functional combines horizon area, Hawking
temperature, local scaffold density, and dark-energy
order into a
30 reproductive measure for black holes. We further
integrate two more specific ideas from the attached
brief: a
31 Popławski-style Einstein-Cartan bounce as one
candidate interior mechanism, and a time-integrated
Cosmic Fertility Index
32 favoring universes that efficiently produce black
holes. The resulting paper is not a statement of
accepted physics; it
33 is a disciplined conjecture designed to remain
observationally admissible while generating explicit
tests in black-hole
34 spectroscopy, dark-matter environment studies,
cosmological history, and analog gravity experiments.

35

36 ****Keywords:**** black holes; baby universes;
cosmological natural selection; Einstein-Cartan
gravity; dark matter; dark
37 energy; black-hole thermodynamics; speculative
cosmology

38

39 ---

40

41 # 1. Introduction and scope

42

43 Modern cosmology is empirically described by a nearly
flat Λ CDM model with cold dark matter and late-time
44 expansion [1]. Black holes are likewise supported by
multiple classes of observation, including horizon-
scale imaging of
45 M87* and Sgr A* [2,3] and strong-field gravitational-
wave tests that remain broadly consistent with
general relativity
46 and Kerr black holes [4]. The microscopic natures of
dark matter and dark energy, however, remain unknown
. That
47 ignorance leaves conceptual room for constrained
speculation, provided that any new interpretation
reduces to the
48 observed background expansion, large-scale structure
, and compact-object phenomenology in the appropriate
limit.

49
50 The attached Grok brief contributes three major
elements to the present synthesis: (i) a more
explicit yolk or nutrient
51 metaphor for dark matter; (ii) a specific Popławski-
style Einstein-Cartan torsion bounce as a candidate
interior
52 mechanism for daughter-universe birth; and (iii) a
time-integrated “Cosmic Fertility Index” that weights
black-hole
53 production against dark-sector conditions [B0]. Those
elements are incorporated here, but only after being
separated
54 into observationally constrained pieces and
interpretive pieces. This is crucial because some
claims in the brief are
55 stronger than what current evidence warrants.

56
57 The purpose of this manuscript is therefore twofold:
first, to present a more formal journal-like version
of the
58 black-holes-as-eggs hypothesis; second, to state
plainly where the merged superset contains divergence

58 , contradiction,
 59 or underdetermination. The result is less
 rhetorically absolute than the brief, but more
 scientifically disciplined.

60
 61 ---
 62

63 # 2. Integrated source synthesis and points of
 divergence

64

65 Table 1 records the main elements contributed by the
 attached brief and the manner in which they are
 merged into the

66 present paper.

67

68 | Element from attached brief

| Merged treatment in this manuscript

|

Divergence / caution

|

69 |-----|-----
 -----|-----

 -----|-----

 -----|

70 | Popławski-style torsion bounce inside black holes
| Retained
 as one concrete interior mechanism alongside more
 agnostic bounce / pinch-off / white-hole transition
 possibilities. | Not adopted as the
 unique mechanism; it remains a candidate among
 several speculative quantum-gravity completions
 .

- 70 |
- 71 | Dark matter as gestational “yolk” or nutrient mass
 |
 Generalized into a scaffold component ρ_{sc} that still behaves approximately as clustering pressureless matter. | Literal nutrient language is metaphorical. The physical requirement is only that the component remain observationally close to cold dark matter
 . |
- 72 | Dark energy as a consciousness / organizing intelligence
 |
 Represented mathematically by a scalar-field order parameter ϕ with $w \approx -1$ in the Λ CDM limit. | The consciousness reading is explicitly interpretive, not empirically established. The safer physics statement is “effective order field
 .” |
- 73 | Cosmic Fertility Index F
 |
 Merged with a local fecundity functional into a single hierarchical reproductive measure F_{tot}
 . | The weighting function is phenomenological and not derived from a completed theory of quantum gravity
 . |
- 74 | Black-hole information paradox resolved by transfer of the genome to the daughter universe |
 Recast as a conjectural inheritance channel via horizon information capacity
 . |
 This is not a solved paradox in accepted physics; it is a proposed route, not a demonstrated resolution
 . |
- 75 | Current acceleration parameter quoted as $w_0 = -1.03 \pm 0.03$ | Softened to

```

75 "$w$ is observationally consistent with  $w = -1$  to
    current precision, with some combined analyses
    exploring mild evolution." | The stronger numerical
    statement is dataset- and model-dependent; Planck
    base  $\Lambda$ CDM itself assumes  $w = -1$  [1], while later
    combined analyses test departures from that value [5
    ]. |
76
77 ---
78
79 # 3. Observational boundary conditions
80
81 Any viable egg hypothesis must preserve four
    empirical boundary conditions.
82
83 ### (i) The background universe must obey the
    Friedmann dynamics of an approximately flat
    Friedmann-Lemaître-Robertson-Walker spacetime:
84
85 $$
86 H^2(t) = \frac{8\pi G}{3} \rho_{\mathrm{tot}}(t) - \frac{k}{a^2(t)}
87 $$
88
89 with
90
91 $$
92 \rho_{\mathrm{tot}} = \rho_b + \rho_r + \rho_{\mathrm{dm}} + \rho_{\mathrm{de}}.
93 $$
94
95 ### (ii) The dominant dark-matter component must
    continue to behave, to a good approximation, like
    clustering pressureless matter:
96
97 $$
98 w_{\mathrm{dm}} = \frac{p_{\mathrm{dm}}}{\rho_{\mathrm{dm}} c^2} \approx 0,
99 \quad
100 c_{s,\mathrm{dm}}^2 \approx 0.
101 $$

```

```

102
103 ### (iii) The dark-energy sector must remain close
      to a smooth accelerating component, with  $w$  near  $-1$ 
      at current precision [1,5].
104
105 $$
106 w_{\mathrm{de}} = \frac{p_{\mathrm{de}}}{\rho_{\mathrm{de}} c^2} \approx -1.
107 $$
108
109 ### (iv) Black-hole exteriors must remain
      approximately Kerr-like at current observational
      accuracy [2-4].
110
111 ---
112
113 # 4. Statement of the egg hypothesis
114
115 We now state the merged hypothesis in formal
      postulate form.
116
117 **P1.** The universe is a living cosmological system
      in the minimal operational sense that it stores
      information,
118 processes free energy, and participates in lineage-
      like propagation.
119
120 **P2.** Black holes are reproductive structures (“
      eggs”) whose interiors either directly seed daughter
      universes or
121 realize intermediate stages in a longer reproductive
      chain.
122
123 **P3.** Dark matter contains, or may be interpreted
      as containing, a scaffold component that feeds or
      stabilizes egg
124 formation while remaining observationally close to
      cold dark matter.
125
126 **P4.** Dark energy may be represented by an order
      field that coordinates large-scale reproductive

```

126 dynamics; in its
 127 strongest metaphysical reading, this field may be
 interpreted as consciousness-like, but that
 interpretation is not
 128 required by the mathematics.
 129
 130 ****P5.**** Universes that generate black holes
 efficiently are statistically favored in a lineage
 or selection picture, in
 131 the sense developed by Smolin [6,7].
 132
 133 ---
 134
 135 # 5. Mathematical framework
 136
 137 ## 5.1 Exterior black-hole geometry and horizon
 thermodynamics
 138
 139 For a non-rotating black hole of mass M , the
 Schwarzschild radius is
 140
 141 $r_s = \frac{2GM}{c^2}$,
 142 $r_s = \frac{2GM}{c^2}$,
 143 $r_s = \frac{2GM}{c^2}$,
 144
 145 and the horizon area is
 146
 147 $A = 4\pi r_s^2 = \frac{16\pi G^2 M^2}{c^4}$.
 148 $A = 4\pi r_s^2 = \frac{16\pi G^2 M^2}{c^4}$.
 149 $A = 4\pi r_s^2 = \frac{16\pi G^2 M^2}{c^4}$.
 150
 151 Standard semiclassical thermodynamics then assigns
 the Bekenstein-Hawking entropy [8] and Hawking
 temperature [9]:
 152
 153 $S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar}$,
 154 $S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar}$,
 155 $S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar}$,
 156
 157 $T_H = \frac{\hbar c^3}{8\pi G M k_B}$.
 158 $T_H = \frac{\hbar c^3}{8\pi G M k_B}$.

159 \$\$
 160
 161 The attached brief treated the horizon entropy as a
 genetic blueprint transmitted to a daughter universe
 [B0]. In the
 162 merged manuscript, that statement is retained as a
 conjectural interpretation rather than a theorem.
 What is established
 163 is only that horizon area scales the thermodynamic
 information capacity of the black hole.
 164
 165 ## 5.2 Interior cosmology and a candidate torsion-
 bounce mechanism
 166
 167 The attached brief specified a Popławski-style
 Einstein-Cartan torsion bounce as the concrete route
 by which a daughter
 168 spacetime emerges from a black hole interior [B0].
 We incorporate that proposal as one member of a
 broader mechanism
 169 class. In the Einstein-Cartan picture, spin-torsion
 coupling can generate short-distance gravitational
 repulsion strong
 170 enough to avoid a singularity and replace it with a
 bounce [10]. The interior daughter domain is then
 modeled by the
 171 Friedmann equation
 172
 173 \$\$
 174 \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k c^2}{a^2} + \frac{\Lambda c^2}{3}.
 175 \$\$
 176
 177 Here ρ may be decomposed as
 178
 179 \$\$
 180 \rho = \rho_b + \rho_r + \rho_{\mathrm{dm}} + \rho_{\mathrm{sc}},
 181 \$\$
 182
 183 where ρ_{sc} is a scaffold component

183 introduced for the present hypothesis. The merged
 paper remains
 184 agnostic as to whether the true mechanism is torsion
 -induced bounce, quantum pinch-off, white-hole
 transition, or some
 185 more general topology-changing event.
 186
 187 ## 5.3 Dark-sector parameterization
 188
 189 To preserve observational compatibility, the
 scaffold component must gravitate similarly to cold
 dark matter on large
 190 scales. In the Newtonian regime we therefore require
 191
 192 \$\$
 193 \nabla^2 \Phi = 4\pi G \left(\rho_b + \rho_{\mathrm{cdm}} + \rho_{\mathrm{sc}} \right).
 194 \$\$
 195
 196 For dark energy we introduce a scalar order field ϕ
 ϕ , mathematically identical to a standard
 quintessence-like
 197 representation:
 198
 199 \$\$
 200 \rho_\phi = \frac{1}{2} \dot{\phi}^2 + V(\phi),
 201 \$\$
 202
 203 \$\$
 204 p_\phi = \frac{1}{2} \dot{\phi}^2 - V(\phi),
 205 \$\$
 206
 207 \$\$
 208 w_\phi = \frac{p_\phi}{\rho_\phi}.
 209 \$\$
 210
 211 When $\dot{\phi}^2 \ll V(\phi)$, one has $w_\phi \approx -1$, recovering the Λ CDM limit. The
 212 consciousness-language of the brief is then recast
 as an interpretive layer on top of an otherwise
 standard effective

213 field description.

214

215 ## 5.4 Combined fertility functional

216

217 The attached brief proposed a time-integrated Cosmic Fertility Index [B0], whereas the earlier draft used a local

218 black-hole fecundity functional. The natural superset is a nested quantity in which local black-hole fecundity is first

219 defined and then integrated across the cosmic history of black-hole formation.

220

221 Define the local egg-competence weight of a black hole i by

222

223 \$\$

224 $\mathcal{F}_i =$

225 $\left(\frac{A_i}{4\ell_P^2}\right)^\eta$

226 $\left(\frac{\rho_{\mathrm{sc}}(x_i)}{\bar{\rho}_{\mathrm{dm}}}\right)^\alpha$

227 $\exp\left[-\beta \frac{T_{H,i}}{T_*}\right]$

228 $\left(1+\xi \frac{V(\phi)}{\rho_{\mathrm{crit}}} c^2\right).$

229 \$\$

230

231 Then define the global, time-integrated fertility index by

232

233 \$\$

234 $F_{\mathrm{tot}}(t_0)$

235 =

236 $\int_{t_0}^{t_0} dt$

237 $\int dM, d^3x,$

238 $\dot{n}_{\mathrm{BH}}(M,x,t),$

239 $\mathcal{F}(M,x,t).$

240 \$\$

241

242 A daughter-domain nucleation rate may be parameterized as

243

244 \$\$
 245 $\Gamma_i = \Gamma_0 \exp\left[-I_E(\mathcal{F}_i)\right],$
 246 \$\$
 247
 248 with a threshold-like Euclidean barrier, for example
 249
 250 \$\$
 251 $I_E(\mathcal{F}_i) = I_0 \max\left(0, 1 - \frac{\mathcal{F}_i}{\mathcal{F}_c}\right).$
 252 \$\$
 253
 254 In words: larger horizon area, lower Hawking
 temperature, denser scaffold environment, and
 stronger order-field
 255 contribution all increase reproductive competence.
 The exact form is phenomenological, but it makes the
 hypothesis
 256 testable.
 257
 258 ## 5.5 Formal propositions and corollary
 259
 260 ### Proposition 1 (External compatibility)
 261
 262 If all new gestational or reproductive physics is
 confined to horizon interiors or is only weakly
 coupled to the
 263 exterior, then the black-holes-as-eggs hypothesis
 can be parameterized so as to remain observationally
 consistent with
 264 current black-hole imaging and gravitational-wave
 tests.
 265
 266 ****Proof sketch.**** The current data constrain the
 exterior geometry far more strongly than the
 interior completion.
 267 Horizon imaging and ringdown analyses are compatible
 with standard black-hole solutions, so any
 admissible egg model
 268 must approach the GR exterior as a limiting case.
 269

270 **### Proposition 2 (Area-temperature fecundity scaling)**

271

272 Within any egg model that weights reproductive potential positively with horizon information capacity and negatively

273 with Hawking temperature, larger and colder black holes are generically more fecund than smaller and hotter ones.

274

275 ****Proof sketch.**** Since $A \propto M^2$ while $T_H \propto M^{-1}$, any monotone fecundity functional $F(A, T_H, \dots)$ that increases with A and decreases with T_H will favor high-mass black holes. This conclusion

276 does not require a detailed interior theory.

277

278 **### Proposition 3 (Λ CDM limit)**

279

280 The merged egg hypothesis contains standard Λ CDM as an observational limit when ρ_{sc} is dynamically

281 indistinguishable from cold dark matter and ϕ behaves as a slowly varying or effectively constant vacuum-energy

282 field.

283

284 ****Proof sketch.**** Set $\rho_{\mathrm{sc}} \rightarrow 0$ or absorb it into cold dark matter, and let $\dot{\phi}^2 / V(\phi) \rightarrow 0$. The background equations then reduce to the ordinary flat Λ CDM form used in current cosmology.

285

286 **### Corollary 1 (Best egg candidates)**

287

288 If the merged fertility functional is approximately correct, the most reproductively competent black holes are expected

289 to be old, massive, cold black holes embedded in dense dark-sector environments.

290

291 ****Proof sketch.**** This follows directly from
Proposition 2 together with the scaffold-density and
order-field factors

292 in $\mathcal{F}_i$.

293

294 ---

295

296 # 6. Mechanistic branches: direct and indirect
reproduction

297

298 The attached brief distinguished direct and indirect
reproduction [B0]. The merged version preserves
that distinction.

299

300 ****Direct branch.**** A single black hole seeds a
daughter universe with slightly altered low-energy
parameters, in the

301 spirit of cosmological natural selection [6,7].

302

303 ****Indirect branch.**** A black hole is one step in a
chain: for example, stellar collapse leads to
massive black-hole

304 growth, mergers reorganize the egg population, and
only some later stage in the chain triggers daughter
-domain

305 formation.

306

307 The present manuscript does not choose between these
branches. Rather, it observes that both fit the egg
metaphor, and

308 both remain largely unconstrained because no
accepted theory yet fixes the fate of black-hole
interiors.

309

310 ---

311

312 # 7. Experimental and observational program

313

314 The value of the hypothesis lies less in its
metaphysics than in the tests it motivates. Several
were introduced in the

315 attached brief and are retained here in more
explicitly graded form.

316

317 ## 7.1 Precision ringdown spectroscopy

318

319 Use current and next-generation gravitational-wave
detectors to search for extremely small, environment
-dependent

320 residuals in quasinormal-mode spectra or echo-like
structures. Current LVK analyses find no evidence
for large

321 deviations from GR or for post-merger echoes [4], so
any positive signal in this framework must be
subtle rather than

322 dramatic.

323

324 ## 7.2 Horizon imaging and black-hole polarimetry

325

326 Use EHT-like imaging and polarimetry to test for
weak, population-level deviations correlated with
environment,

327 accretion history, or host dark-matter content. The
working expectation is not gross shadow distortion-
already

328 constrained by M87* and Sgr A* [2,3]-but tiny
systematic trends.

329

330 ## 7.3 Dark-matter density profiles around candidate
eggs

331

332 Search for unusually strong dark-sector
concentrations around dormant or weakly accreting
massive black holes. This is

333 the observational version of the attached brief's
nutrient-flow idea [B0]. Stellar proper-motion
studies, maser-disk

334 kinematics, precision lensing, and future dark-
matter searches are relevant here.

335

336 ## 7.4 Cosmological correlation between horizon
inventory and dark-energy history

337

338 Estimate a comoving horizon-area density $\rho_{A_H(z)}$
from black-hole mass functions, merger rates, and
AGN demographics,
339 and compare its growth to reconstructions of dark-
energy evolution. A significant correlation would
not prove the
340 hypothesis but would support a dynamical link
between reproductive investment and cosmic
acceleration.

341

342 ## 7.5 CMB inheritance tests

343

344 The attached brief proposed searching for parent-
universe imprints in CMB non-Gaussianities or
polarization
345 anomalies [B0]. This remains highly speculative, but
it can be formalized as a search for statistically
unusual
346 correlations between large-angle anomalies and the
later cosmic black-hole population.

347

348 ## 7.6 Analog-gravity and quantum-simulation studies

349

350 Laboratory analog horizons in Bose-Einstein
condensates, fluids, or optical systems cannot
create universes, but they
351 can test whether horizon-like systems exhibit
threshold transitions in information flow, effective
bounce behavior, or
352 state transfer. The attached brief rightly
emphasized that these are currently feasible as
theory-discriminating
353 analogs [B0].

354

355 ## 7.7 Theoretical inheritance mapping in parameter
space

356

357 Following Smolin's logic, simulate neighboring
universes with small changes in constants and ask
whether black-hole

358 productivity or integrated fertility decreases. If
our universe is near a local maximum in reproductive
capacity, that

359 would count as circumstantial support [6,7].

360

361 ---

362

363 # 8. Discussion

364

365 The strongest form of the hypothesis treats the
universe as literally alive and dark energy as
literally conscious. The

366 present manuscript permits that reading but does not
require it. The weaker and more scientifically
disciplined form

367 says only this: the dark sector may encode presently
unknown degrees of freedom whose effective behavior
stabilizes

368 structure formation, accelerates expansion, and
perhaps couples to black-hole interiors in a lineage
-like way. That

369 weaker form is still radical, but it is less
vulnerable to immediate dismissal.

370

371 A second caution concerns the information paradox.
The attached brief stated that the paradox is
resolved because

372 horizon information becomes the genome of the
daughter universe [B0]. In the merged paper this is
softened. Horizon

373 entropy does make the inheritance metaphor
mathematically suggestive, but an actual derivation
of unitary information

374 transfer is not supplied here and remains absent
from accepted theory.

375

376 A third caution concerns overclaiming consistency.
The model can be made consistent with current
observations by

377 construction, but no current dataset requires it.
That distinction matters. This is a viable

377 speculative framework, not
378 an empirically favored one.
379
380 ---
381
382 # 9. Conclusion
383
384 By combining the earlier draft with the attached
Grok brief, we obtain a sharper and more explicit
version of the
385 black-holes-as-eggs idea. The merged manuscript
preserves the useful elements of the brief—Popławski
-style torsion
386 bounce, the Cosmic Fertility Index, and a concrete
experimental program—while clearly marking its
stronger claims as
387 conjectural rather than established. The result is a
disciplined speculative cosmology in which the
universe functions
388 as a living lineage-bearing system, black holes act
as eggs, dark matter may serve as gestational
scaffold, and dark
389 energy may be treated as an order field with room
for a stronger metaphysical interpretation. Its
scientific value
390 depends entirely on whether future observations
reveal the predicted links among black-hole
demographics, horizon
391 physics, dark-sector environments, and cosmic
expansion history.
392
393 ---
394
395 # References
396
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Hypothesis for Reproductive Cosmology in a Living
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M87 Event Horizon Telescope Results. I. The Shadow
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    Replicating Universes in
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431
432 ---
433
434 # Appendix A. BibTeX bibliography
435
436 The following entries are provided in BibTeX form
    for reuse.
437
438 ```
439 @misc{GrokEggBrief2026,
440 author      = {{Grok 4.2 Beta}},
441 title       = {Black Holes as Cosmic Eggs: A
    Hypothesis for Reproductive Cosmology in a Living
    Universe},
442 year        = {2026},
443 note        = {Uploaded markdown brief provided by
    the user; incorporated into the present synthesis}
444 }
445
446 @article{Planck2020CosmologicalParameters,
447 author      = {{Planck Collaboration} and Aghanim,
    N. and others},
448 title       = {Planck 2018 results. VI.
    Cosmological parameters},

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449 journal      = {Astronomy \& Astrophysics},
450 volume       = {641},
451 pages        = {A6},
452 year         = {2020},
453 doi          = {10.1051/0004-6361/201833910},
454 eprint       = {1807.06209},
455 archivePrefix= {arXiv},
456 primaryClass = {astro-ph.CO}
457 }
458
459 @article{EHT2019M87Shadow,
460 author        = {{Event Horizon Telescope
    Collaboration} and Akiyama, K. and others},
461 title         = {First {M87} Event Horizon Telescope
    Results. I. The Shadow of the Supermassive Black
    Hole},
462 journal      = {Astrophysical Journal Letters},
463 volume       = {875},
464 pages        = {L1},
465 year         = {2019},
466 doi          = {10.3847/2041-8213/ab0ec7},
467 eprint       = {1906.11238},
468 archivePrefix= {arXiv},
469 primaryClass = {astro-ph.GA}
470 }
471
472 @article{EHT2022SgrAShadow,
473 author        = {{Event Horizon Telescope
    Collaboration} and Akiyama, K. and others},
474 title         = {First Sagittarius A* Event Horizon
    Telescope Results. I. The Shadow of the Supermassive
    Black Hole in the Center of the Milky Way},
475 journal      = {Astrophysical Journal Letters},
476 volume       = {930},
477 number       = {2},
478 pages        = {L12},
479 year         = {2022},
480 doi          = {10.3847/2041-8213/ac6674}
481 }
482
483 @article{Abbott2021TestingGRGUTC2,

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484 author      = {Abbott, R. and others and {{LIGO
    Scientific Collaboration}} and {{Virgo Collaboration
    }}},
485 title       = {Tests of General Relativity with
    Binary Black Holes from the Second LIGO--Virgo
    Gravitational-Wave Transient Catalog},
486 journal     = {Physical Review D},
487 volume      = {103},
488 pages       = {122002},
489 year        = {2021},
490 doi         = {10.1103/PhysRevD.103.122002},
491 eprint       = {2010.14529},
492 archivePrefix= {arXiv},
493 primaryClass = {gr-qc}
494 }
495
496 @misc{DESI2025DR2Guide,
497 author      = {{DESI Collaboration}},
498 title       = {DESI DR2 Results: March 19 Guide},
499 year        = {2025},
500 howpublished = {Lawrence Berkeley National
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